

- #4 Repeat this procedure for each of the solutes in turn. Where the solute is a liquid add about 0.5 mL to the kerosene.
- #5 Discard the contents of the test tubes down the sink and clean the test tubes thoroughly.
- #6 Place 2–3 mL of distilled water into the cleaned test tubes and test the solubility of each of the solutes in water. Record your results.
- #7 Discard the contents of the test tubes down the sink and clean as before.

Processing of results, and questions

- 1 What generalisations can you make about the solubility in water and kerosene of
 - a ionic substances
 - b polar covalent molecular substances
 - c non-polar covalent molecular substances?
- 2 If two pieces of clothing were stained with chromium(III) chloride and lubricating grease respectively, what types of solvent would be needed to remove the stains?

Experiment

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Solubility of nitrates, chlorides, sulfates and carbonates of some common cations

Background

Ionic substances dissolve in water to varying extents. The different solubilities of these substances have several applications. For example, chemists use a knowledge of solubilities to identify the presence of certain ions, both cations and anions, and also to prepare compounds.

In this experiment the solubilities of nitrates, chlorides, sulfates and carbonates will be examined. This will be done by observing the results of mixing solutions of the sodium salts of the above mentioned anions with solutions containing different cations.

Equipment required

Test tubes (several)

Test tube rack

Solutions of the following (all 0.1 mol L⁻¹, about 20 mL):

sodium nitrate [NaNO₃]
 sodium chloride [NaCl]
 sodium sulfate [Na₂SO₄]
 sodium carbonate [Na₂CO₃]

Solutions of the following in dropper bottles (all 0.1 mol L⁻¹):

potassium nitrate [KNO₃]
 silver nitrate [AgNO₃]
 lead(II) nitrate [Pb(NO₃)₂]
 ammonium nitrate [NH₄NO₃]
 magnesium sulfate [MgSO₄]
 iron(II) sulfate [FeSO₄]
 copper(II) sulfate [CuSO₄]
 zinc sulfate [ZnSO₄]
 chromium(III) sulfate [Cr₂(SO₄)₃]
 aluminium sulfate [Al₂(SO₄)₃]
 barium chloride [BaCl₂]
 calcium chloride [CaCl₂]
 iron(III) chloride [FeCl₃]

Beakers (two 100 mL)

Droppers (two)

Procedure

#1 Draw up a results table as shown below.

Solution to be tested	Cation present	NaNO ₃	NaCl	Na ₂ SO ₄	Na ₂ CO ₃
		Anion added			
		NO ₃ ⁻	Cl ⁻	SO ₄ ²⁻	CO ₃ ²⁻
KNO ₃	K ⁺	n.r.	n.r.		
AgNO ₃	Ag ⁺	n.r.	white ppt.		
Pb(NO ₃) ₂	Pb ²⁺				
NH ₄ NO ₃	NH ₄ ⁺				
MgSO ₄	Mg ²⁺				

- #2 Place about 1 mL of each of the solutions to be tested (KNO₃, AgNO₃, Pb(NO₃)₂ etc) into separate clean test tubes.
- #3 Place about 20 mL of 0.1 mol L⁻¹ NaNO₃ into a clean 100 mL beaker. Use a dropper to add about 1 mL of the NaNO₃ solution to each of the solutions in the test tubes. Where a precipitate occurs indicate its colour in the table. If no reaction is observed indicate with the letters 'n.r.'
- #4 When all of the solutions have been tested with the NaNO₃ solution carefully clean the test tubes, and repeat the procedure with the 0.1 mol L⁻¹ NaCl solution (column 2), 0.1 mol L⁻¹ Na₂SO₄ solution (column 3) and 0.1 mol L⁻¹ Na₂CO₃ solution (column 4).

Processing of results, and questions

- Examine your observations in column one of the results table. Can any generalisation be made about the solubilities of nitrates?
- Examine columns two, three and four of the table in the same way.
- Write ionic equations for all cases in which a precipitate was formed.

Experiment

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Identification of unknowns

Background

The purpose of this investigation is to identify a set of unknown solids or solutions.

Your teacher will allocate you one or more of the following sets of solids or solutions. Each member of the set provided will be labelled only as A, B, C ... etc.

Set 1: 1 mol L⁻¹ solutions of BaCl₂, CuSO₄, H₂SO₄, NaCl randomly labelled as A, B, C, D.

Set 2: 1 mol L⁻¹ solutions of AgNO₃, ZnSO₄, BaCl₂, NaI randomly labelled as G, H, I, J.

Set 3: 1 mol L⁻¹ solutions of H₂SO₄, BaCl₂, Na₂CO₃, Pb(NO₃)₂, KOH and 2 mol L⁻¹ HCl randomly labelled as M, N, O, P, Q, R.

Set 4: Solid samples of BaCl₂, Zn(NO₃)₂, Na₂CO₃, CaCO₃ randomly labelled as W, X, Y, Z. (Distilled water, dilute hydrochloric acid and limewater solution are also provided to test the unknown solid samples.)

It will be necessary for you to plan a procedure to identify the unknowns provided, carry out the procedure and evaluate your results.

Equipment required

Unknown samples in Set 1, 2, 3 or 4 as listed above with each chemical labelled only as A, B, C etc.

Set 1–3 solutions in dropper bottles

Set 4 solid samples in small sample bottles; if Set 4 unknowns are used, the following solutions in labelled dropper bottles are also provided:

Hydrochloric acid (HCl) – 1 mol L⁻¹

Limewater solution (Ca(OH)₂)

Spatula

Test tube racks

Test tubes (several)

Distilled water

Beakers (two 100 mL)

Planning the investigation

- Plan an investigation to identify the unknown solids or solutions provided. For Sets 1–3 you may only use the solutions provided; for Set 4 the solid samples can be tested with the hydrochloric acid, limewater and distilled water provided. For each test, use a minimum amount of sample—for solids about a pea-sized sample; for solutions about 2–3 mL.